CEN 4010 Team Project – Iteration 0

Note: Requirements list may be expanded to include additional customization features as the remaining time for the project allows (i.e. it is probably incomplete).

1. Problem Statement:
   The software will provide detailed feedback to help piano learners become more proficient in sight-reading piano sheet music without a teacher.
2. Project Scope:
   This desktop software will generate random measures of sheet music that are displayed to the user through the software's GUI. It will use algorithms to generate "pianistic" passages following common structures that piano students might see in real pieces they are studying. It will detect user input through a piano keyboard MIDI controller (and a computer keyboard for testing purposes). The user's input will be compared with the generated sheet music, and an accuracy score will be calculated based on how closely the user followed the sheet music. The user can then choose to generate another set of random measures and try again as many times as he or she pleases and get an average accuracy calculated across all attempts. The software will not receive audio input or produce audio output, nor will it contain functionality for printing or editing music sheets directly.
3. Functional Requirements:
   - On application startup, a screen containing a title and a "Start" button is displayed to the user.
   - When the user clicks the "Start" button, the following are displayed to the user:
     - Three measures of randomly generated sheet music in the 4/4 time signature.
     - A gray horizontal bar located at the bottom of the application.
     - An attempt counter initialized at 0.
     - A "Reset Attempt Counter" button.
   - When the user enters a valid MIDI input from a connected MIDI controller or a computer keyboard, the gray horizontal bar becomes red.
   - When the user enters a number of valid MIDI inputs that is greater than or equal to the number of randomly generated music notes, the following actions take place:
     - The red horizontal bar becomes blue.
     - An accuracy score for the attempt is displayed to the user.
     - An average accuracy score across all attempts is displayed to the user.
     - A "Retry" button is displayed to the user.

- When the user clicks the “Retry” button, the following actions take place:
    - Three measures of randomly generated sheet music in the 4/4 time signature are displayed to the user.
    - The blue horizontal bar becomes gray.
    - The “Retry” button disappears.
    - The accuracy score for the attempt disappears.
    - The average accuracy score across all attempts disappears.
- When the user clicks the “Reset Attempt Counter” button, the following actions take place:
    - The attempt counter is set to 0.
    - The average accuracy score across all attempts is reset to 0 and disappears, if visible.

4. <u>Non-functional Requirements:</u>
    - A physical MIDI controller is required to develop and test parts of the system.
    - The software must be developed within the remaining time left in the semester (about 10 weeks).
    - For a smoother user experience, the software’s latency between button presses and their corresponding actions should be less than 100 milliseconds.
    - The software should function as expected with zero failures or crashes.